63

$SSD_1 = 0.278 \times 90 \times 2.5 + \frac{V^2}{254(f \pm n)}$ → brake efficiency 50%

$= 0.278 \times 90 \times 2.5 + \frac{90^2}{254 \times 0.7 \times 0.5}$

$= 153.6\ m$

$SSD_2 = 0.278 \times 60 \times 2.5 + \frac{60^2}{254 \times 0.7 \times 0.5} = 82.19\ m$

$SSD = SSD_1 + SSD_2 = 82 + 153$

$= 235.8\ m.$

Q find SSD for descending gradient of 2% for a design speed 80 kmph velocity given $22.22\ m/s^2$ coefficient of longitudinal friction = 0.35

Ans $SSD = vt_R + \frac{v^2}{2g(f \pm N/100)}$

$SSD = 22.22 \times 2.5 + \frac{22.22^2}{2 \times 9.81 \left[(0.35 \times 1) - \frac{2}{100}\right]}$

$SSD = 131.9\ m.$

(a) What is ISD

Ans @ $ISD = 2SSD$

$= 131.9 \times 2$

$= 264\ m$

Traffic engineering

traffic study

- Basic objective
  - to
  - fac
  - pvmt
  - acc
- units
  - traffic vol
  - traffic s
  - traffic speed
  - traffic cap

Part 2 Traffic engineering & pavement design / Highway part 2

Traffic engineering (Chapter 1)

Traffic study

The basic objective of traffic study

- to obtain knowledge of type and volume of traffic at present, in future
- To study road facility & improvements to be done
- to design geometric features & pavement thickness
- to analyse accidents

(1) Traffic volume (vehicle per hour)

(2) Traffic density (vehicle per km)

(3) traffic speed (kmph)

(4) traffic capacity = $\frac{\text{traffic volume}}{\text{lane}}$

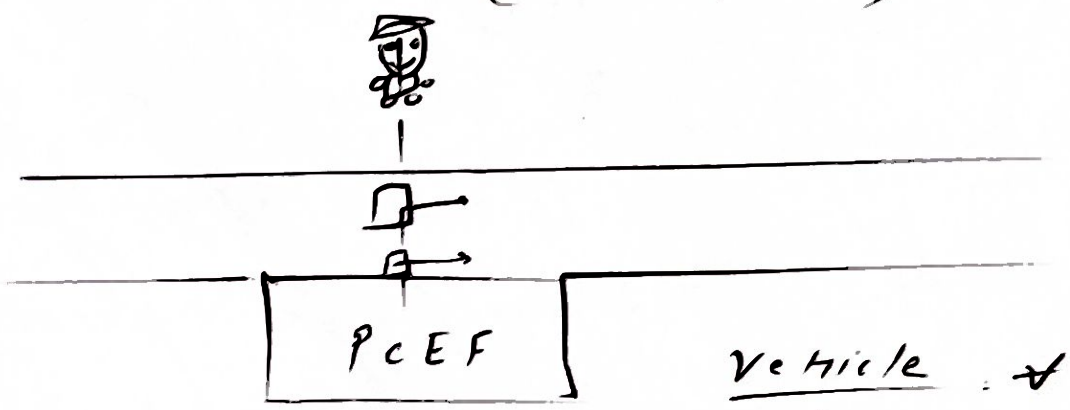
$= \frac{\text{vehicle/hr}}{\text{lane}}$

= vehicle/hr/lane

6.5) traffic vol
- alternate name
- expressed as
- unit

traffic volume / traffic flow

is expressed as the number of vehicle crossing a section of road per unit time. (vehicle/hour)



Vehicle / hour : 4

7:00am – 8:00am

- study
  - counting mthds
    - man
    - mech
      - all name
      - interes.
    - mech counting tools
      - PN
      - P3
      - VC
      - bend
      - μm

Traffic volume study

1) Counting methods
   1. manual traffic counting
   2. mechanical counting / automatic counted
   3. traffic volume counted @ intersecns

Mechanical counting is done by -
1) Pneumatic tube
2) piezoelectric cable
3) Videocamera
4) Bending plate.
5) Micro mm wave radar detector

Traffic volume Study

- Presentn of data
  - ① AADT
    - kyah?
  - ② ADT
    - kyah?
  - ③ Trend chart
    - kyah?
  - ④ Traffic flow map
    - alternate names
    - use?

Presentation of traffic vol data

① AADT (Average annual daily traffic volume)

The average 24 hr traffic volume at a given location over full 365 days of year is called AADT.

② ADT (average daily traffic volume)

It is the twenty four hr average traffic volume @ given location for some period of time less than 1 year.

③ Trend chart

Showing traffic volume trend over perid of years.

④ Traffic flow map or diagram along routes

Generally used to represent traffic @, @ intersections

6.2 Traffic study

- Present methods
  - ⑤ 30th highest hourly vol
    - kya hai
    - kya design krte h isse
    - congestion okkab hota hai
  - ⑥ Peak hr factor
    - kya hai
    - kis chez ka measure h?
    - formula

⑤ 30th highest hourly vol

The hourly volume that will be exceeded only 29 times in a year & all other hourly volumes of the year will be less than this volume

① Highway facilities are generally designed for this volume.

② Congestion only during 29th hour in the year.

⑥ Peak hour factor PHF

. The hourly volume during the maximum volume hour of the day divided by 15 minute flow rate within the peak hour is called peak hour factor.

. It is a measure of traffic demand fluctuations within peak hour.

. PHF : $\frac{1}{4}$ $\frac{q}{q_{15}}$ or $\frac{1^{th}}{4}$ of { dinka wo hour jisme traffic max / Us hour ke max 15 min ka traffic

presentation TVS

PHF

- PHF for highway

a The resultant of traffic count taken between 5 pm to 6pm find out peak hr for for this freeway also find out the actual (design) flow rate for the actual traffic volume

time interval	traffic vol.
5:00-5:15	900
5:15-6:30	1000
5:30-5:45	1200
5:45-6:00 pm	850

★ very imp ???

q is peak hour volume (vehicle/hr) (5.8

q_{25} volume during the peak 15 minute of flow. (unit vehicle/15 min) or (vehicle/$\frac{1}{4}^{th}$ hr)

for highway PHF = 0.8 - 0.95

Ans $PHF = \frac{1}{4} \times \frac{q}{q_{15}}$

q = peak hour volume (vehicle/hour)

= 900 + 1000 + 1200 + 850

= 3950 vehicle/hour.

q_{15} = 1200 vehicle/15 min.

$PHF = \frac{1}{4} \frac{3950}{1200} = 0.823$

Actual design flow rate = $\frac{q}{PHF}$

$= \frac{3950}{0.823} = 4800$ vehicle/hr.

7.1

Speed Study Class1 10 Aug

Speed study
- kyo kartehi
- Methods
  - ① eno/mirr
    - Mirrors kisanglepe set krtehi
    - enoscope distance frm observer
  - ② Other
    - a) electric
    - b) spt gn
    - c) elect m+r
    - d) time lapse

- Speed study is required to analyse highway geometrics.

- Methods

1. Mirror box method or enoscope
   - Mirrors are set @ 45°
   - Generally enoscope is kpt 50m from observer

2. other methods
   a) photoelectric method
   b) Radar gun / spot gun → Spot speed
   c) , , electronic meter mthd
   d) time lapse method

- Uses of spotspd
  - 1) plng trfc
  - 2) accs
  - 3) geo des
  - 4) trafcap

- Uses of spot speed study
  1) In planning & regulating traffic
  2) In accident studies
  3) In highway geometric design
  4) to calculate traffic capacit

11.4 Q2022 the thickness design of pavement mainly depends on -

- Q CBR Test
  - kiske liye hai
  - kya hai
  - lab kiske liye hai
  - cylinder based
  - plunger Q
  - Std. values kab lete h
  - Test.

Ans design wheel load

CBR Test -

is done for soil sample

is a measure of resistance to penetration of soil or flexi. pavement material of standard plunger under controlled test condns

it is a lab method h.

cylinder mold base = 150 mm ϕ

plunger diameter = 50 mm ϕ

The standard load values are taken @ 2.5mm & 5mm penetrn. & then CBR values calculated.

Test.

TEST

mould me soil sample lete h aur sasp plate 2.5 mm tak penetration depth se load ki value nikaalte hain ... (1).

(CBR) $_{2.5mm\ penetrn} = \frac{P_1}{\text{Std load}} \times 100\%$.

— " — $= \frac{P_1}{1370\ kg} \times 100\%$. (load formula)

$= \frac{P_1\ kg}{70\ \frac{kg/cm^2}{kg/cm^2}} \times 100$ (pressure formula)

iske baad penetrn depth 5 mm lete h. aur load value P_2 find out krte h.

$CBR_{5mm\ penetrn} = \frac{P_2 / \text{Std load}}{2055\ kg} \times 100\%$. (load formula)

$= \frac{P_2\ kg}{105\ kg/cm^2} \times 100$ (pressure formula)

.it is fo0d that $CBR_{2.5mm}$ is always more than CBR_{5mm}

Hence CBR considered : $CBR_{2.5\ mm}$

. agar CBR_5 ki value $CBR_{2.5}$ se zyada ata to is case me test should be repeated after repeating the test if CBR_5 again comes more than $CBR_{2.5}$ then consider ans as CBR_5.

Test
- 2.5 mm wala kaise lete h
  - formula normal
    - pressure
- 5 mm wala
  - formula normal
    - pressure
- CBR 2.5 & 5 ka relation & CBR considern
- exception al case me kya karenge.

11.c)

Standard load Value on crushed Stone.

Table

Penetrn mm	Std load kg	Stdstress kg/cm²
2.5mm	1370 kg	70 kg/cm²
5 mm	2055 kg	105 kg/cm²

for CBR test material passing 20mm sieve is only considered.

kis material ko CBR test te lie consider karenge

Q imp ESE2016 A collapsible (Soil subgrade sample was tested using standard CBR test the outcomes are given as

Sno	load kg	Penetrn
1	60.55 kg	2.5mm
2	80.55 kg	5mm

find out CBR val of sample.

Ans:

Sno	load	Penetrn	Std	CBR
1	60.55	2.5 mm	1370	$\frac{60.55}{1370} \times 100 = 4.4\%$
2	80.55	5.0 mm	2055	$\frac{80.55}{2055} \times 100$ $= 3.39\%$

CBR = 4.4%

q ESE 2017: in Revised CBR design method recommended by IRC for the design of flexible pavement the total thickness depends on -

ESE 2011

Q CBR method which statements are correct

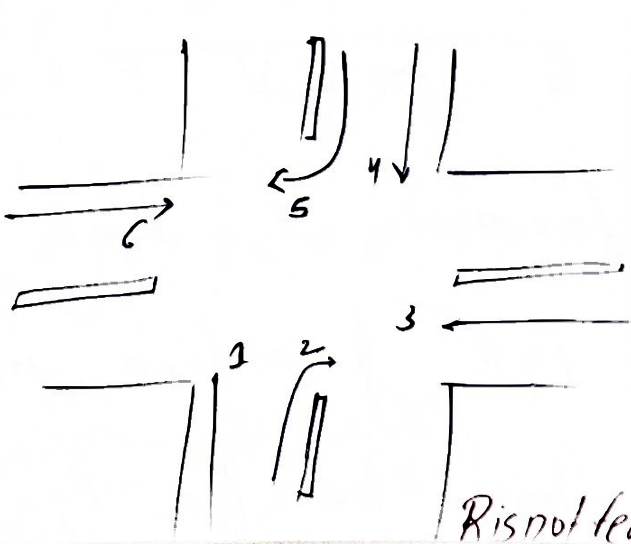
1) CBR test is conducted insitu
2) Static compression is best adopted
3) The top 50 cm layer of subgrade should be compacted to as nearal proctor density

The CBR value of soil &
cumollative standard axle load.

Ans 1) wrong CBR test is a lab test
2) correct.
3) Correct.

---

q) which one of the following combinns is not possible



	I	II	III
a P	1,4	2,5	3,6
b Q	1,2	4,5	3,6
c R	2,5	1,3 fail	4,6 fail
d S	1,4	2,6	3,5

Ans

R is not feasible, because

11.8 ESE 2015 Q What is the initial traffic after condition if annual average daily traffic count is 400 - CVPD, traffic growth rate per annum is 7%. The road is proposed to be completed in 3 years

Ans: $A = P\left[1 + \frac{7}{100}\right]^n$

$= 400\left[1 + \frac{7}{100}\right]^3$

$= 400\left[1 + 0.07\right]^3$

$= 400 \times 1.07^3$

$= 490$ CVPD

Q What will be the theoretical maximum capacity for a single lane of highway given that the speed of traffic stream is 40 kmph

Ans: $C = 1000 \frac{V}{S}$

$S = SSD + 6.1$ — (1)

$= 0.278 V t_R + \frac{V^2}{254 g f} + 6.1$

alternate formula

$S = 0.2V + 6$ — (2)

formula 1 tab use karenge jab question me ssd diya hoga nai toh eqn 2 use karenge

S = 0.2 × 40 + 6

= 14 m

C = 1000 × 40/14

= 28 57.14 vehicle / hr

= 2960.

Q. International traffic intelligent system

Classified traffic volume study

survey data are related to Q5 16, the normal flows on two approach roads at an inter Section are resp. 500 pcu/hr & 300 pcu/hr. The corresponding saturation flow is 1600 pcu/hr on each road. total lost time per signal cycle is 16 seconds then optimum signal cycle time as per webster's method is ...

Ans 18. $N_1 = 500$, $N_2 = 300$ pcu/hr

$S_1 = 1600$ pcu/hr, $S_2 = 1600$ pcu/hr

$L = 16$ seconds

$C_o = \frac{1.5L + 5}{1 - y} = \frac{1.5 \times 16 + 5}{1 - 0.5}$ = 58 sec. Ans

$y = y_1 + y_2 = 0.99 \simeq .5$

$y_1 = \frac{N_1}{S_2}$ $y_2 = \frac{N_2}{S_2}$

$= 500/1600 = 0.3125$ $= 300/1600 = 0.1875$

148

Class 2 16 Aug 2024.

- Pavement materials
  - ① Soil tests/Subgrade
    - ⓐ CBR
    - ⓑ GI
      - formula
        - a
        - b
        - c
        - d
        - f
        - wl
        - Ip
      - Range
    - ⓒ PLT
      - insitu or: lab
      - Seating load val
      - Seating pressure
      - plate dia

Pavement materials & test:

1) Subgrade or soil test.

a) CBR

b) GI.

formula $G = 0.2a + 0.005ac + 0.01bd$

$a = F - 35$

$b = F - 15$

$c = w_l - 40$

$d = I_p - 10$

F is %age finer than 75 μ sieve

w_l is liquid limit

I_p is plasticity index

Range 0 to 20, 0 best 20 worst

c) Modulus of subgrade rxn determin by plate load test (in situ)

- Seating load = 320 kg
- Seating pressure = 0.07 kg/cm²
- plate diameter = 75 ø cm

Soil test

- PLT
  - formulas
    - Determination of k
      - (i) standard settlement wala
      - (ii) Determination of k Standard pressure wala

k is ... kg/cm³

or N/mm³

Determination of k

formulas (i) Standard settlement

t = 1.25 mm

$$k = \frac{P \text{ kg/cm}^2}{(1.25 \times 10^{-1}) \text{ cm}} \quad : \quad \frac{P \text{ kg/cm}^2}{0.125 \text{ cm}}$$

$$\Rightarrow k = \frac{P}{0.125} \frac{kg}{cm^3}$$

P is pressure applied

(ii) Standard pressure

p = 0.7 kg/cm²

$$k = \frac{0.7 \text{ kg/cm}^2}{\delta \text{ cm}}$$

$$k = \frac{0.7}{\delta} \text{ kg/cm}^3$$

12.3

Soil tests
- Plate load test
  - formulas
    - Determin of k
      - std load
      - std press
    - Determin of E of soil
      - flexi plate formula
    - for rigid plate formula
    - flexi plate me E/s zyada milta h ki rigid plate
    - Modulus of subgrade rxn of soil aur sample kadigah toh kaise krege

Determination of modulus of elasticity of soil

(i) for flexible plate

$$E_s = \frac{1.5\, P_{max}}{\delta}$$

E_s → soil, kg/cm²

modulus of elasticity = 1.5 × pressure on plax / maximum vertical deflection @ centre of flexi plate × radius of plate

kg/cm² * cm / cm

(ii) for rigid plate

$$E_s = \frac{1.18\, P_q}{\delta}$$

if modulus of subgrade reaction of soil & any other sample is given then

	k_s	P_s
1 soil sample	k_1 ,	P_1
2 Sample	k_2 ,	P_2

Pavement materials

- (a) test on soil material subgrade

$k \propto P$

$\frac{k_1}{k_2} : \frac{P_1}{P_2}$

$\frac{k_s}{k_2} : \frac{P_s}{P_2}$

$\Rightarrow \boxed{\frac{k_s}{P_s} \times P_2 : k_2}$

- (b) Test on aggregate
  - i. Toughness test
    - Impact value test
      - Impact kyah?
    - toughness kyah?
  - no of blows
  - height of fall
  - wt of hammer
  - no of layers
  - formula

Tests on aggregate

(a) aggregate impact value (Toughness)

Impact is resistance of aggregate against impact loading

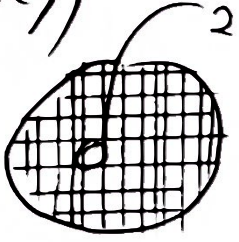
Alternate name toughness

no of blows = 15

height of fall = 38 cm.

hammer wt = 14 kg (or 15 kg)

3 layers of aggregate



Aggregate impact value = $\frac{\text{Weight of aggregate passing 2.36 mm sieve}}{\text{total wt of aggregate}} \times 100\%$

12.5

Pavement mat
- Tests on soil.
- Tests on aggregate
  - Impact value
    - higher value kya dikhati h?
    - wearing course impact value
    - bituminous macadam AIV
    - WMBM [base course]
  - strength test
  - crushing val
    - kya hota h crushing val?
    - no of layers
    - no of blows

More will be aggregate impact value then more chances of aggregate to break into small pieces (poor sample). lesser the will be aggregate impact value then less possibility of aggregate to break. (good sample)

for wearing course AIV ≯ 30%.

for bituminous macadam layer AIV ≯ 35%

for WMBM [base course] AIV ≯ 40%.

strength test

crushing value test

it is the resistance against gradual load [gradual compressive load]

no of layers = 3

no of blows = 25

Crushing value test

formula

aggregate crushing value

$$= \frac{\text{weight of aggregate passing through 2.36 mm sieve}}{\text{total wt of aggregate}} \times 100\%$$

- max allowable crushing value
  - Surface or wearing course
    - for base course

- Surface wearing coose crushg value ≯ 30%.
- Base course acv ≯ 45%.

Tests on aggregate

- Boundness test
  - kya h?
  - Sample ko oven dry temp
  - kaunsa soln dalte h
  - Water absorptn

Boundness test. [Durability test]

- Resistance of aggregate to weathering action.
- Sample ko oven dry krte h 105-110°c
- Solution of Sodium sulphate or magnesium sulphate is used.

Water absorption test

12.7 Tests on aggregate

- wabsrpn
  - maxval
- Specificgrav test
  - acc the aggregate ke liye range
- Shapetest
  - Names (3) of tests
  - kaunse sequence me tests karte?

Water absorption ∴ 0.6% of weight of sample

Specific gravity test

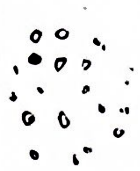
- Range for good road aggregate 2.6 to 2.9

Shape test

1) flakiness Index test
2) Elongn Index test
3) Lularity number test.

flakiness & elongation tests are done in sequence.

flakiness is performed first & then in selected - non flaky - particles elongation index is done.



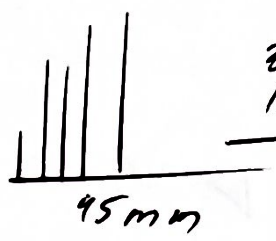
100 Nos → mean size 25mm → fineness test → $\frac{3}{5} \times 25 = 15mm$ → less pass → 32 nos flaky particles → $FI = \frac{32}{100} \times 100$

$= 32\%$

$100 - 32 = 68$ non flaky particles

Elongn index

68 nos → $1.8 \times 25 = 45$



37 nos length graded → $EI = \frac{37}{68} \times 100\%$

$= 54.4\%$

Combined index: $FI + EI = 32 + 54.4$

$= 86.4\%$

Q Given a test result in 500gm sample ould which 100gm is flaky particle & 80 gram sample are elongated then what will be flakiness index, elongation index combined index.

Ans: 500gm → flaky 100gm, non flaky = 500 − 100 = 400gm; Elongated 80gm

$FI = \frac{\text{flaky particles wt}}{\text{total wt}} \times 100\%$

$= \frac{100gm}{500gm} \times 100\% = 20\%$

$EI = \frac{\text{Elongated particle wt}}{\text{non flaky particle wt}} \times 100\%$

$= \frac{80gm}{400gm} \times 100\% = 20\%$

Combined Index: $FI + EI$

$20\% + 20\%$

$= 40\%$

12.9

tests of shape

- flakiness kya test
  - size ke krnye sample size
- Angularity no
  - kyah?
  - kyaqus dikhata hy?
- formula
- Range?

for flakiness & elongation particle size & sample size selected should be more than 6.3 mm.

Angularity Number is an index or measure of voids in excess of rounded gravel for which void ratio is 33%.

it also shows reduction in roundness

An = V_v - 33%.

= $\left[67\% - \left(\frac{100W}{CG}\right)\%\right]$ [solid formula]

V_v is volume of voids in %age

W is mean weight of aggregate in cylinder (gm).

C is weight of water required to fill the cylinder gm

G is specific gravity of aggregate 2.6 to 2.9

pavement materials

- Tests on Aggregate
  - Abraision test:
    - Kya hai
    - types?
    - Abraision vs attrition.
  - LA test
    - RPM, drum radius
    - formula
    - max value or range
      - for wearing course
      - for base course

Range of An 0 to 11
Good bad

is always represented in nearest whole no Eg 1.2 ≃ 2

Abraision test: [Hardness test]

Resistance to wear or abraision or scratch or hardness.

1. Los angeles abrasion test
2. Deval's abrasion test
3. Dorry's attrition test

Abraision means rubbing action between different particles

Attrition means rubbing action between same particles

Los angeles abrasion test.

Rpm 30 to 33, drum radius 350 mm

formula: $\frac{\text{wt of aggregate passing through 1.7mm sieve}}{\text{total weight}} \times 100\%$

los angle abrasion value ↓ LAV

	AIV	ACV	LAV
wearing	⩽30	⩽30	⩽30% ← for wearing course
bituminous macadam	⩽35%		
base course	⩽40%	⩽45%	

LA abrasion for base course of bituminous bound macadam ⩽50% ← macadam.

for cc construction LAV ⩽16%.

Class 13 17 Aug 2024

Conversion / Gradient to percent kena.

x % = 6 %

$\frac{6}{100} = \frac{6/6}{100/6} : \frac{1}{16.67}$

∴ 1 in 16.67

Bitumen → Tar

use of cutback ?

- Cutback — Grade 0 to 5 (least / most viscous)
  - used in colder region
  - surface dressing
  - soil stabilisation
- emulsion
- asphalt

RC cutback kahan use hota hai

Rapid curing cutback is used in patch work

MC cutback fluxed with ?

Medium curing cutback is fluxed with kerosene or diesel oil

SC cutback fluxed with furnace oil

Tar

emulsion : bitumen suspended in finely divided aqueous medium, used in wet weather

Tar

- RT 1 use

RT 1 (road tar grade 1)

- used for surface painting in cold weather with very low viscocity

- RT 2 use

RT 2

- Recommended for standard surface painting under normal climatic condition.

- RT 3 use

RT 3

- used for surface painting renewal coat & premixing chips.

- RT 4 use

RT 4

- used for premixing tarmacadam in base courses

- RT 5 use

RT 5

- used for grouting purpose

13 Asphalt

- kya ko ta h?

Asphalt

- If the bitumen contains some inert material of mineral then it is called asphalt

Bitumen tests

- Penetration tests
  - kya determine kre ta h?
    - Determination of hardness & softness is done by penetration test
  - Machine.
    - Penetrometer
  - unit of measurement.
    - it can be measured by depth tenth of millimeter $\frac{1}{10}^{th}$ mm penetration
  - isse kya krte ta?
    - grading of bitumin is done by this test value.
  - General value
    - 80/100 the range of 80 to 100
    - ② standard test condition, needle penetration 8mm to 10mm.
  - Low penetration value
    - 30/40 or 3mm to 4mm ← lowest penetration value

Tests on bitumen

1. Penetration test (25°C)

Bitumen tests

- Ductility test
  - kya hoti h ductility
  - unit & test temp.
  - kis chez ka measure h
  - instrument
- Viscocity test
  - kya h viscocity
  - alternate name.
  - instrument
  - test temp
  - representh.

Ductility test

. Ductility is the property by which material (bitumen) can be converted or stretched into wires

. It is expressed in cm. & test temperature 27°C with 50 mm per min pull.

. This test is measure of adhesiveness and elasticity of bitumen.

. standard briquette setup

3 Viscocity test

. Viscocity is a measure of resistance to flow.

. It is also called consistency.

. orifice type viscometer

. test temp 25°

. represent in time or no of seconds required for 50 ml of material to flow through an orifice.

Bitumen tests

(13:17)

- viscocity test
  - time viscocity rel"
- Softening pt test
  - instrument
  - gives result in
  - Softening pt defn or kya hota h Softening pt.
  - range
  - High softening pt & temp kya rel" h?

. More will be the time required, more is the viscocity

Softening point test

. Ring & ball apparatus

. Gives result in temperature

. it is the temperature @ which bitumen or substance attains a particular degree of softening under specified condition of test.

. usual range 35° to 70°c

. higher the softening pt, lower will be temperature susceptibility & is preferable in warm climate.

- Specific gravity
  - instrument
  - SG Pure bit
  - SG tar

Specific gravity test

Pycnometer

Pure bitumen SG 0.97 to 1.02

tar SG 1.10 to 1.25